

Beyond Aggregate Success: Paired Statewise Evaluation of VLA Policy Updates

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Abstract: Quantized, distilled or otherwise accelerated vision-language-action (VLA) policies are judged by the change in their aggregate success rate. A deployment also needs to know whether the new policy fails where the old one succeeded. Comparing the two policies with one rollout per initial state cannot distinguish an update effect from rollout variability: in VLAQuantBench’s public LIBERO records, rerunning an unchanged configuration with a new seed changes the outcome of up to a quarter of the initial states. We define a state as a LIBERO initial state together with its scene seed, repeat each state under both policies with shared policy-seed plans, and use exact paired tests with false-discovery control. On X-VLA in LIBERO-10, the first-rollout old-versus-old floor is 2–3 flips per 100 states. Three-bit weight rounding concentrates a 20-point aggregate drop on a few tasks; seven states change from 5/5 to 0/5 successes, but the five-repeat screen cannot yield a corrected statewise discovery. Four-bit rounding and fewer denoising steps produce no detected statewise regression, which does not certify their absence. BF16 leaves the aggregate nearly flat but identifies one candidate state; 20 fresh paired rollouts confirm its loss (18/20 against 4/20, exact $p = 2.59 \times 10^{-4}$). This is evidence about one independently retested state, rather than an estimate of regression prevalence or a physical safety certificate.

Keywords: Vision-language-action models, quantization, evaluation, regression testing

1 Introduction

VLA updates change precision, inference steps or checkpoints. Aggregate success rates [1] do not answer an operator’s question: where does the new policy fail when the validated old one succeeded? The question resembles backward compatibility in classification [2, 3, 4], where a more accurate model can still err on inputs the old one got right. Closed-loop control adds rollout randomness and evolving observations, which complicate the interpretation of an individual outcome change.

On LIBERO [5], Sawada and Kasahara [6] find substantial outcome changes under compound perturbations; Yardımcı and Çoğurcu [7] report task collapse under online RL. An observed old-success/new-failure pair may be an update-induced loss, a stochastic fluctuation, or a change in the scene. These explanations require different responses. A single rollout does not distinguish them, even when aggregate success rises.

Our contributions are an empirical noise baseline from four models and four suites (Section 3), a paired statewise protocol with repeat controls (Section 2), and one fresh confirmation of a BF16 loss in X-VLA [8] (Section 4). We distinguish single-rollout flips, repeated-state candidates and fresh confirmation, using established statistical tools rather than proposing a new test.

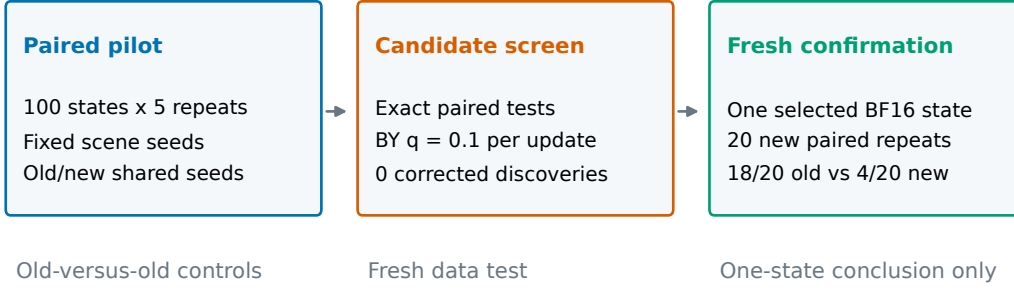


Figure 1: Discovery and confirmation use different outcomes. The 100-state pilot yields descriptive candidates but no BY-corrected discoveries. One selected BF16 candidate is tested on 20 fresh pairs, excluding its pilot outcomes. Old-versus-old controls accompany the pilot; confirmation supports a conclusion about the selected state only.

2 Paired statewise evaluation

State identity. LIBERO provides an initial-state file per task for the robot and movable objects. Fixtures such as cabinets and stoves can nevertheless change when a scene is rebuilt: three hard resets of LIBERO-10 task 0, state 0 gave three cabinet positions. We define a state $s = (\text{task}, \text{initial-state index}, \text{scene seed})$ and use a fixed scene-seed plan for comparisons of policy variants. This identity is an operational evaluation definition, not proof that every hidden simulator variable is identical.

Pairing. Each state is rolled out n times per variant. The policy-noise seed plan depends on the task, batch and base seed, and does not depend on the variant. We pair outcomes by state and repeat index and check that their recorded policy seeds match. Shared seeds can reduce comparison noise, but do not guarantee identical noise consumption or trajectory coupling after the policies diverge. Each historical evaluation batch creates a new vector environment. The archived records do not contain full reset-state or camera digests (Appendix A).

Exact paired test. For state s , let b_s count old-success/new-failure pairs and c_s count old-failure/new-success pairs. We test the one-sided null that harmful discordance is no more likely than beneficial discordance. At the null boundary, conditional on $b_s + c_s$ discordant repeats,

$$p_s = \sum_{k=b_s}^{b_s+c_s} \binom{b_s+c_s}{k} 2^{-(b_s+c_s)}. \quad (1)$$

Set $p_s = 1$ when there is no discordance. This is the conditional exact paired McNemar/binomial test, rather than an unpaired comparison of success counts. Validity assumes independent paired repeats from a stable evaluation process; dependence within each old/new pair is allowed. Pseudo-random seed matching alone does not establish these assumptions.

Multiplicity and descriptive summaries. For each update, we apply Benjamini–Yekutieli (BY) [9] at FDR $q = 0.1$ across its 100 statewise regression hypotheses. BY permits arbitrary dependence between valid statewise p -values. Improvement tests, where reported, form a separate directional family; neither directional result constitutes a joint guarantee over both directions or over all searched updates.

We also report the Jeffreys posterior $\Pr\{\text{Beta}(b_s+1/2, c_s+1/2) > 1/2\}$ for the direction of discordance. This posterior is a descriptive candidate score: it is not a multiplicity-adjusted discovery rule or a posterior on the magnitude of the success loss. Aggregate intervals resample the 100 observed

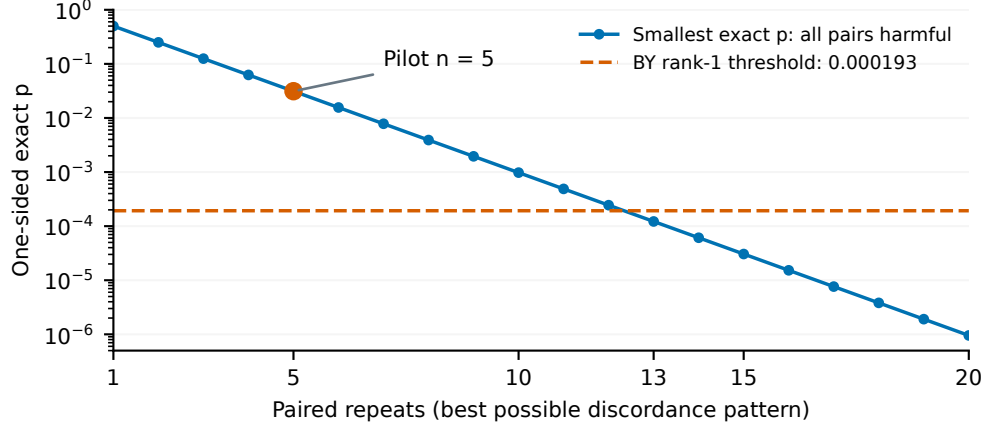


Figure 2: Best-case test resolution. The blue curve assumes every repeat is a harmful discordance; any gain or concordance raises the attainable p -value for a fixed total number of repeats. The orange line is the BY rank-1 cutoff for 100 hypotheses at $q = 0.1$. The pilot cannot produce a corrected discovery. The crossing at 13 repeats is an arithmetic bound under this extreme outcome pattern, not a power analysis.

states 4,000 times with a fixed analysis seed and use the 2.5th and 97.5th percentiles of their mean paired differences. They are descriptive state-bootstrap intervals, conditional on this inventory and its observed repeats, rather than a guarantee for unseen tasks or states. The *naive* flip count uses repeat 0 only.

Old-versus-old controls. We compare a policy with itself using new policy seeds, pairing planned repeat indices. A separate control changes scene seeds and pairs task/initial-state indices. That control is a scene perturbation, not an identical-state policy update. Both measure repeat variability; they do not establish that an individual update-induced loss is absent.

Why discovery needs a second stage. With $n = 5$, the smallest attainable one-sided exact p -value is $2^{-5} = 1/32$, even if every pair is harmful. For 100 states, the BY rank-1 cutoff is $q/(100H_{100}) \approx 1.93 \times 10^{-4}$, with $H_{100} = \sum_{i=1}^{100} 1/i$. Even the rank-100 cutoff, $q/H_{100} \approx 0.0193$, is below $1/32$. Thus no state can pass this pilot’s BY rule, regardless of its outcome pattern. This is a limitation of the design, not evidence that the updates preserve every state. Figure 2 shows the attainable resolution. Thirteen entirely harmful pairs would cross the rank-1 cutoff, but this best-case calculation is not a general sample-size or power recommendation.

3 How noisy is one rollout per state?

VLAQuantBench [1] released episode records from its 409 closed-loop runs under an MIT licence. Its LIBERO configurations cover four models and four suites. Each configuration uses 200 states (20 initial states for each of 10 tasks), except X-VLA, which uses 190. Twenty configurations are repeated with additional seeds. We reanalyze release commit 4a2cb7c; these are public benchmark outcomes, separate from our X-VLA experiments.

Unchanged configurations flip many states. Table 1 and Figure 3 show outcome changes between seeds of the same configuration. Near the ceiling, the count is small (1–8 of 200 at 97.5–99.5% success), but rises as success falls: 21–32 states at 72–88% and 32–50 at 60–71%. The plot includes all released baseline-versus-update comparisons and all 20 repeated configurations; representative rows appear in the table.

Table 1: Representative unchanged-configuration reruns in VLAQuantBench (200 states). Success rates and flips are raw descriptive counts. “Never/always/mixed” partitions states across all listed seeds. The reanalysis records contain all 20 repeated configurations.

Suite/model	Configuration	Success by seed	Pair flips	Never/always/mixed
Object/ $\pi_{0.5}$	baseline	.990 .985 .995	5, 1, 4	0/195/5
Object/ $\pi_{0.5}$	W4A4, action head	.600 .630 .610	44, 50, 50	43/85/72
Spatial/ $\pi_{0.5}$	W4A4, action head	.705 .705 .685	44, 36, 32	34/110/56
Spatial/ $\pi_{0.5}$	W4A4, MLP only	.880 .875	29	10/161/29
10/OpenVLA-OFT	W3, no fc2 in head	.935 .915 .910	8, 7, 9	11/177/12

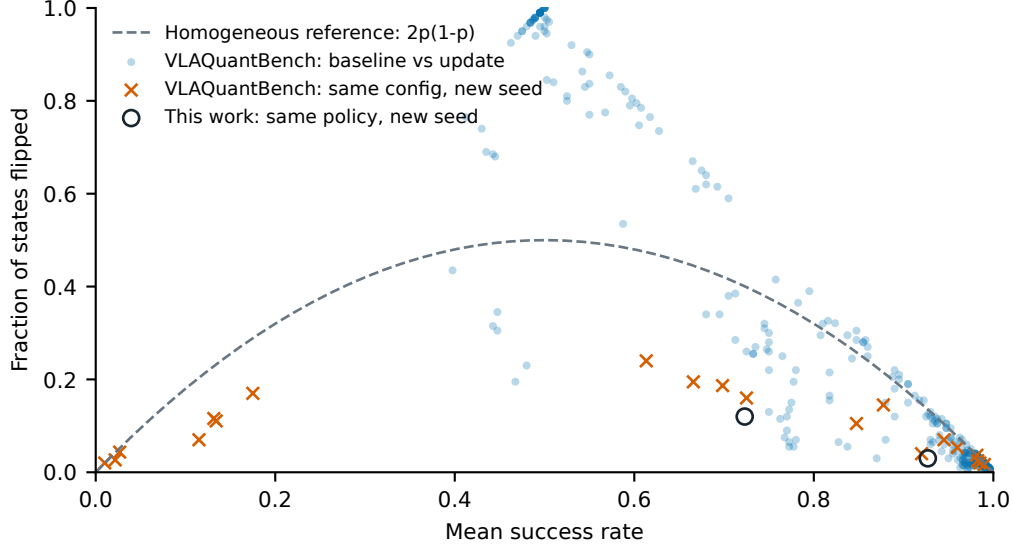


Figure 3: Outcome flips versus mean success. Blue dots: public baseline-versus-update comparisons. Orange crosses: unchanged public configurations with new seeds (mean over available seed pairs). Open circles: our FP32 and W3 policy-seed controls, first repeat only. The dashed homogeneous-state reference $2p(1 - p)$ assumes two independent rollouts with common success probability p ; it is not a fitted noise model or a confidence bound.

Equal seed labels do not establish pairing. Different configurations run with the same seed disagree on as many states as with different seeds. For example, $\pi_{0.5}$ baseline against its W4A4 action-head variant on LIBERO-Object differs at 77.3 states with matching seed labels and 78.0 across seeds (means over the available comparisons). The released labels do not by themselves demonstrate episode-level common-random-number coupling. We use reruns to contextualize the observed flips, without assigning the public cross-configuration flips an exact paired significance interpretation.

State difficulty is heterogeneous. For $\pi_{0.5}$ with a W4A4 action head on LIBERO-Object (61% success), 43 states fail under all three seeds and 85 succeed under all three. If every state instead had the same success probability, independence would predict approximately 12 and 46. This contrast supports heterogeneity in state difficulty; it does not identify a causal update loss from one rollout.

Consequence for a small aggregate change. For π_0 [10] on LIBERO-10 with W8 weights, aggregate success moves from 48.5% to 47.5%, while 24 states flip to failure and 22 to success. A seed change alone produces a comparable flip count at that success level. Interpreting the locations of those update flips requires repeated outcomes.

Table 2: FP32 versus each variant: 100 states, 5 repeats per state. Δ and descriptive 95% state-bootstrap intervals are in percentage points. Naive flips use repeat 0 (harm/gain). BY counts use separate regression/improvement families at $q = 0.1$. Posterior counts are descriptive. FP32 baseline success is .926.

Variant	Success	Δ [95% CI]	Naive	BY	$P(\text{worse}) > 0.9$
FP32, new policy seeds	.928	+0.2 [−1.0, +1.4]	0/3	0/0	0
FP32, new scene seeds	.930	+0.4 [−0.4, +1.2]	0/2	0/0	0
W4, group 128	.918	−0.8 [−3.2, +1.2]	4/2	0/0	1
W3, group 128	.726	−20.0 [−26.8, −13.4]	23/2	0/0	26
2 flow steps (of 10)	.932	+0.6 [−0.6, +2.0]	1/3	0/0	0
BF16	.916	−1.0 [−4.6, +2.2]	4/3	0/0	5

4 X-VLA precision and inference-step updates

Setup. We evaluate `lerobot/xvla-libero` using LeRobot 0.6.1 [11] on LIBERO-10: 10 tasks $\times$ 10 initial states $\times$ 5 repeats, absolute end-effector control, 10 flow-matching steps, and a 520-step limit. Every failure in the FP32 run is a timeout. Our baseline succeeds in 92.6% of episodes; VLAQuantBench uses a 900-step horizon and reports 97.9% for the same model identifier. These protocols differ, so their rates are context rather than a controlled replication.

The W3 and W4 variants round all 267 linear layers (792M parameters) to a symmetric integer grid with one scale per group of 128 inputs, then store the weights dequantized. They measure a numerical weight perturbation, not the speed or memory efficiency of an integer inference kernel. The two-step variant changes the flow-matching step count; BF16 changes model numerical precision.

The floor is low for high-success fixed states. FP32 against itself changes 13 of 500 outcomes under new policy seeds and 8 under new scene seeds. No state’s success count changes by more than two. Of the 100 states, 88 succeed in all five FP32 repeats and five fail in all five. At this success level and under this protocol, reruns are close to replay. The scene-seed control does not show a larger flip count than the policy-seed control, but this small comparison does not establish that fixture placement is irrelevant. At lower success, W3 against itself changes 73 of 500 outcomes. Its first repeats give 3 harmful and 9 beneficial flips; four states have posterior direction probability above 0.9, and none passes BY.

W4: descriptive candidates, no corrected discovery. Its six naive flips are twice the first-repeat policy-seed floor. The most affected state changes from 5/5 to 1/5 (paired $p = 0.0625$). No state passes correction, as expected from the screen’s resolution limit. This is not evidence of equivalence or preservation at every state.

W3: large losses concentrated in particular states. The aggregate drop is 20 points, but its location is uneven. Tasks 5, 8 and 9 retain 96–100% success, while tasks 0, 4 and 7 lose 40–52 points. Seven states change from 5/5 to 0/5. None passes correction; 26 have posterior direction probability above 0.9, compared with four in the W3 rerun control. The first-repeat protocol reports 23 harmful and two beneficial flips. Every W3 failure times out. Its successful episodes average 265 steps, compared with 256 for FP32. VLAQuantBench’s W3 variant loses only 3.2 points under its quantizer and 900-step horizon. Quantizer and horizon differ together, so the source of that between-protocol difference cannot be isolated here.

Two flow steps: higher success with one naive new failure. Reducing the flow-matching head from 10 to two steps changes success from 92.6% to 93.2%. The first repeat still shows one new failure; that state changes from 5/5 to 4/5. This is compatible with the observed repeat floor and is not a corrected statewise discovery. The descriptive aggregate interval also includes zero.

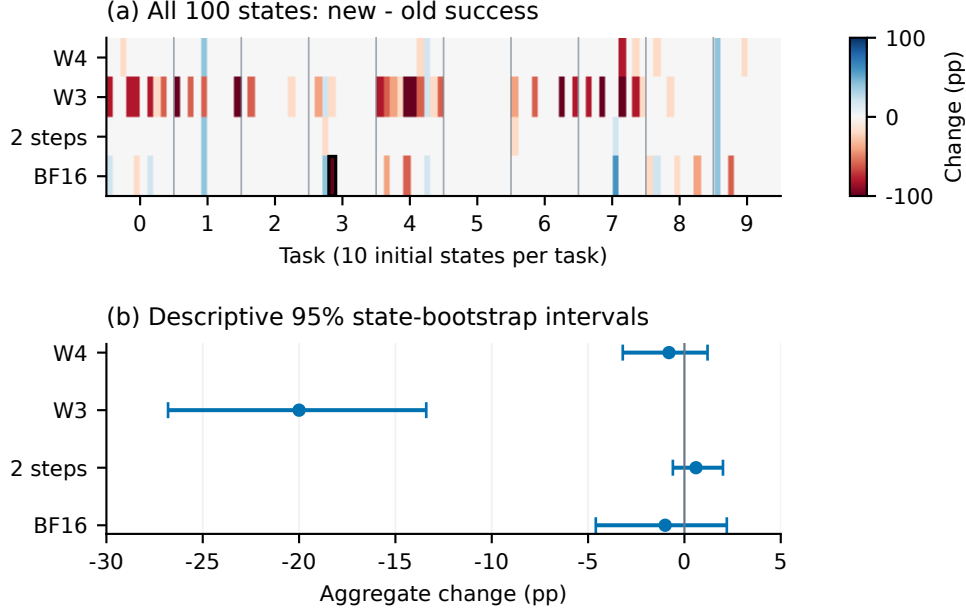


Figure 4: All 100 pilot states in fixed task/initial-state order. (a) Empirical success differences in percentage points, with all five repeats. Black outline: task 3, initial state 3, selected for BF16 follow-up (indices start at zero). (b) Aggregate differences and descriptive state-bootstrap intervals. These locations are candidate patterns: every update has zero BY-corrected discoveries at five repeats.

BF16: one candidate confirmed with fresh pairs. BF16 changes the aggregate by -1.0 points, with a descriptive interval including zero. Recorded episode lengths differ in 422 of 500 pairs; these logs do not provide a full trajectory-distance measure. One state changes from 5/5 to 0/5 (uncorrected paired $p = 0.03125$). The pilot alone does not establish its regression.

We selected this state (task 3, initial state 3) for a targeted follow-up with 20 fresh pairs and a new policy-seed plan. The pilot outcomes are excluded from the test. FP32 succeeds in 18/20 and BF16 in 4/20. The paired table contains three joint successes, 15 harmful discordances, one beneficial discordance and one joint timeout (Figure 5). The one-sided exact probability is

$$p = \Pr\{\text{Binomial}(16, 1/2) \geq 15\} = \frac{\binom{16}{15} + \binom{16}{16}}{2^{16}} = 0.0002593994. \quad (2)$$

The single fresh follow-up provides evidence of a loss at this selected state, conditional on the paired-test assumptions and the evaluation protocol. All 16 BF16 failures are timeouts. It does not estimate how common such losses are across the 100 states, other updates, different horizons or physical robots.

5 Related work and scope

Backward compatibility [2, 3, 4] motivates reporting new errors even when total accuracy improves. In a closed-loop VLA, outcomes depend on action sampling and the evolving environment; therefore a binary flip needs repeat controls before it can be interpreted as a persistent local loss. VLAQuant-Bench [1] measures post-training quantization across models and suites. We reuse its released records to measure repeat variability and add a repeated-state X-VLA study with independent candidate confirmation.

Compound robustness evaluation [6] tests perturbation sensitivity; online-RL task collapse [7] concerns changes during adaptation. Deployment comparisons [12] and update-admission audits [13]

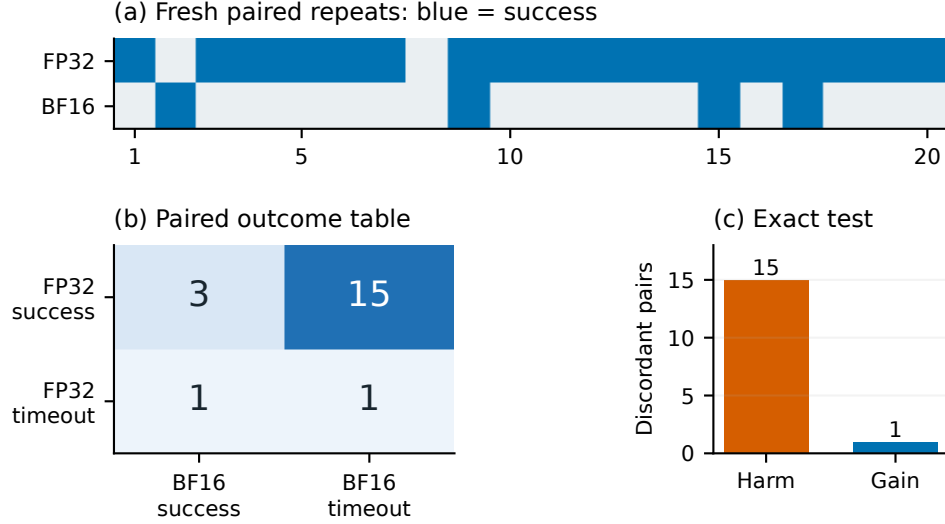


Figure 5: Fresh BF16 confirmation at one selected state. (a) All 20 paired binary outcomes in repeat order; light grey denotes timeout. (b) The full paired table. (c) Fifteen harmful versus one beneficial discordance yield exact one-sided $p = 0.0002593994$. The state was selected from the pilot, but only new outcomes enter this single confirmation test.

examine aggregate behavior or synthetic acceptance settings. Our endpoint is the location of a precision or inference-step update’s candidate loss in a real closed-loop simulation benchmark. Sequential policy comparison [14] suggests allocating additional rollouts to uncertain comparisons. We do not claim a new sequential stopping rule, nor reuse fixed-sample p -values after unrestricted outcome-dependent stopping.

6 Discussion and limitations

Report paired outcomes and old-versus-old controls alongside aggregate success, distinguishing *candidate*, *confirmed loss*, and *not resolved*. The five-repeat screen is incapable of rejecting any statewise hypothesis under its BY rule; a small aggregate change does not overcome that limit.

The new experiments use one model in simulation, a limited fixed inventory and a 520-step timeout criterion. They do not measure physical hazard, unseen-scene robustness, hardware latency or integer-kernel performance. Public-data reanalysis broadens the noise observations, but does not make our precision experiment a multi-model study.

Exact tests require stable independent paired repeats. BY permits dependence across valid state tests; it cannot repair invalid within-state sampling, systematic reset differences or corrupted outcomes. Historical records omit some provenance needed for a fully audited replay. The BF16 follow-up measures a numerical update under this protocol without identifying its internal failure mechanism.

A prospective study should freeze state inventories and update families, validate reset equality, record immutable model/simulator identities, and budget multiplicity and confirmation before collection. Retain timeouts, infrastructure failures and uncertain decisions. These are future improvements, not assertions that unrecorded historical checks were performed.

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A Reproducibility and evidence boundaries

Archived outcomes. The archived research files contain the public-data reanalysis, eight 500-episode pilot/control files, and two 20-episode follow-up files. Rows record task, initial-state index, scene seed, repeat index, policy seed, success, steps, model family, variant and suite. The analysis rejects duplicate repeat identities, unequal state inventories and undeclared policy-seed mismatches. The figure builder checks paired identities, rederives all plotted differences, confirms the follow-up seed separation, and records SHA-256 identities of its inputs. These hashes identify the archived files; they do not prove that missing historical metadata was captured at collection time.

Numerical reanalysis. For a same-scene update, paired rows match (task, initial state, scene seed, repeat). For the policy-seed control, differing policy seeds are explicitly declared. The scene-seed control explicitly drops the scene seed from matching and therefore answers a different question. The bootstrap uses analysis seed 1 and 4,000 state resamples. The analysis implementation includes boundary checks for discordance tests, BY correction and pairing failures.

Historical seed plan. Scene seeds use the declared base, task and initial-state index. Policy random seeds depend on the base, task and batch. The FP32 pilot uses base seed 1000; its independent-policy-seed control uses 2000. The targeted follow-up uses base 3000. Torch TF32 is disabled and deterministic settings are requested with warnings rather than guaranteed deterministic execution. New vector environments are created for each historical batch. Batch composition can affect the consumption of random numbers; changing it is not a documented equivalent replay.

Provenance gaps. The historical episode files do not include immutable checkpoint revisions, complete reset-state/camera digests, or full action and observation streams. The evaluator loads a model by its repository identifier. A currently cached revision cannot establish the exact historical download, so no retrospective checkpoint pin is asserted here. These omissions limit bit-exact reproduction and any stronger causal claim based on hidden simulator-state equality. Later prospective reset audits and health checks are separate work and do not constitute new evidence in this manuscript.

Interpretation checklist. The public-data flips estimate observed repeat disagreement; they are not a statewise significance result. The pilot heatmap and posterior identify candidates; none is a BY-corrected discovery. The fresh BF16 test confirms one selected candidate under its assumptions; it is not an estimate of population prevalence. Non-detection is not equivalence, and simulated weight rounding is not a hardware acceleration measurement.

Local reproducibility. Run `python3 build_figures.py` from this revision directory, then `make all`. The figure builder reads the repository’s archived results. The arXiv upload archive is a flat allowlist of TeX, bibliography output, the unmodified third-party style and the five vector figure PDFs. Its clean-room build does not require the raw data, Python environment or a network connection. The expanded preprint uses the same research outcomes as the workshop submission; it adds exposition, figures and explicit evidence limitations.